Twick o Tweat

Game Script

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GitHub repository: https://github.com/daggerscythe/twick-o-tweat

Engine: Unity 6 (6000.6.4f1)

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1. Game Concept

1.1 Elevator Pitch

Twick o Tweat is a first-person Halloween game where you fill trick-or-treaters' candy orders in a haunted house while shooting the ghosts that float through the walls. Survive from 6 PM to 9 PM and get the best score you can.

1.2 Concept Description

Kids show up at your door with a speech bubble showing which candy they want and how many. You grab a bag, fill it at the candy bowls, and bring it back before the kid's timer runs out, while ghosts float in through the walls and you keep turning around to shoot them with your Ecto-Blaster. Perfect orders earn bonus points, and a perfect order in a bag that matches the kid's color also gives a short power-up. As the night goes on, orders get bigger, a second kid joins the line, new ghost types show up, and finally the power goes out.

1.3 Connection to the Theme

The whole game is one Halloween night from 6 to 9 PM: trick-or-treaters, a haunted house, ghosts, candy, a final phase played with a flashlight during a power outage, and an end-of-night rating in pumpkins.

1.4 Expected Player & Player Experience Goals

Following player-centric design, my representative player is 10 or older, comfortable with WASD and mouse, and likes short, chaotic games like *Overcooked* or *Diner Dash* plus some casual shooting. Using VandenBerghe's Five Domains of Play, they are medium on Novelty (two familiar genres mixed in a new way), medium-high on Challenge (timers, strikes, strict perfect-order bonuses), low on Stimulation (solo play), and low-medium on Threat (spooky, never scary).

The player should feel busy but in control, panic a little when two kids and a tank show up at once, feel clever when they discover the color-match power-ups, and want to play one more night for a better score. I did run into the "I am my own typical player" trap, so things that felt obvious to me, like where the trash can was, weren't obvious to my playtesters, and a lot of the round 1 changes came from that.

2. Genre

Primary genres: **Shooter** (3D, first-person) and **Action and Arcade**, mixed with the order-fulfillment style of *Overcooked*. Unlike most 3D shooters there is no ammo management, because the resource the player is really managing is time.

3. Background Information

3.1 Story & Setting

You just moved into the cheapest house on University Ave in Oxford, and on Halloween you find out why: it's haunted. The house's ghosts were kids who never got to finish trick-or-treating, and

every Halloween they wake up hungry for candy (which is why Thief Ghosts go for your bag). A rumor says the new neighbor gives out the best candy on the block, so the kids keep lining up. You found an old "Ecto-Blaster" in the attic, and your goal is to survive until 9 PM. Most of this backstory is not in the game, since the intro pop-up only says the house is haunted and the ghosts are hungry too.

3.2 Real-World Background

The game is based on real trick-or-treating, which happens roughly between 6 and 9 PM. The five candies are real candies from Kazakhstan (Grilyazh, Kazakhstan, Krokant, Mishutka, and Molochno-Vafelnye).

3.3 What the Player Needs to Know Before Playing

- There is no main menu. The game opens on a paused intro pop-up, so click **Start**. Click in the window to lock the mouse, and press **Esc** to pause and free it.
- Keyboard and mouse only, and sound on is recommended, since audio warns you about ghosts entering behind you and stolen candy.
- There is no save file. Only your high score and volume are saved.

4. What the Game Teaches

4.1 Learning Goals

Twick o Tweat simulates a very busy Halloween night and teaches:

- **Multitasking and prioritizing:** finish the bag or turn around and shoot? Which of two kids first?
- **Time management:** reading green-to-red timers and judging how long a trip to the bowls takes. The faster you deliver, the bigger the speed bonus.
- **Accuracy under pressure:** bonuses only count for exact orders, so you learn to count carefully and use the trash can instead of handing over a wrong bag.
- **Risk vs. reward:** grabbing the matching bag color costs time and only pays off if the order is perfect, but the power-up can save you (Yellow before a Tank, Red when low on health).
- **Spatial awareness and aim:** ghosts come through all four walls, so you learn to check behind you, listen for them, and aim quickly, even in the dark.

In terms of Gardner's intelligences from *Theory of Fun*, the game mostly uses **Logical-mathematical** (counting and scoring), **Spatial** (tracking ghosts in 3D), and **Bodily-kinesthetic** (aim and reaction) skills.

4.2 How Each Concept Is Taught

Concept	Where	How the player is taught
Controls, interacting	Intro pop-up	The intro pop-up lists every control. Looking at anything usable shows a prompt.

Concept	Where	How the player is taught
Bags, bowls, delivering, trash can	Tutorial step 1	One small order with no timer and no ghosts. A hint line reacts to what you do. You need one perfect delivery to move on.
Timers, strikes, shooting, health	Tutorial step 2	A pop-up explains timers, strikes, and ghosts, then one ghost with 1 HP floats in.
Perfect-order bonuses	Phase 1	The phase pop-up gives the tip, and every delivery shows a score breakdown.
Color-match power-ups	Whenever discovered	Hidden on purpose (*Rules of Play*: rules as information). The first perfect order in a matching bag opens a pop-up listing all four effects.
Two kids, Thief Ghosts	Phase 2	A pop-up explains both, and the next ghost is forced to be a thief.
Tank Ghost	Phase 3	A pop-up explains it, and the next ghost is a tank.
Lights Out	Phase 4	A pop-up explains it.

5. Core Mechanics & How to Play

5.1 Controls

Input	Action
W / A / S / D	Move
Mouse	Look / aim
Left click	Shoot the Ecto-Blaster
E	Interact: grab a bag, add a candy, empty the bag at the trash can, give bag
Esc	Pause menu
Enter / Space	Close a pop-up

5.2 Core Mechanics

Kids, bags, and candy
Kids are colored beans standing on the porch, each with a speech bubble showing candy icons and counts and a timer bar (order also shown on the HUD). You grab a bag in any of the four colors, add candy from the candy bowls, and can empty the bag at the trash can between the tables. The bottom left of the HUD shows your bag's color and contents.

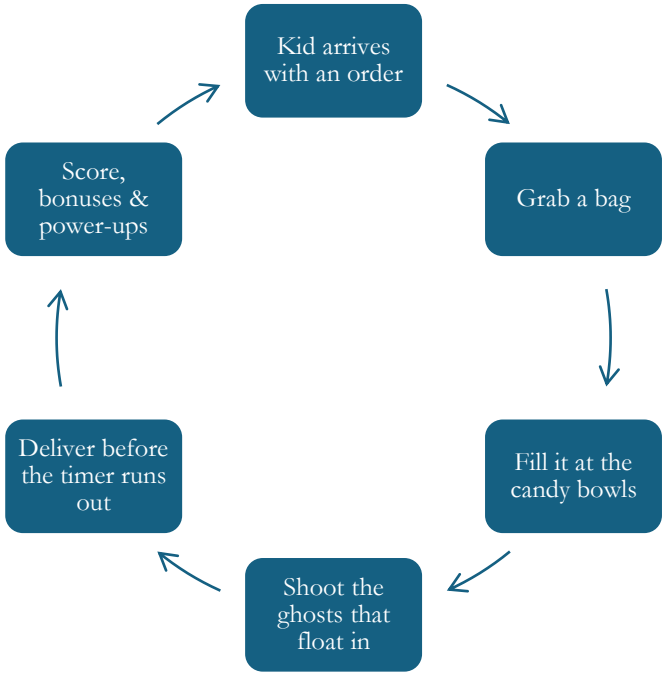
Delivering
Press E at a kid to hand over your bag, even if it's incomplete. Correct candy earns points, but bonuses only come from perfect orders. A perfect order in a bag matching the kid's color also gives

a 5-second power-up: **Light Blue** (faster movement), **Yellow** (double damage), **Red** (instant full heal), or **Green** (ghosts can't hurt you). Different effects stack.

The Ecto-Blaster and ghosts
Left click fires an instant shot with infinite ammo. Shots are blocked by walls even though ghosts float through them, and the gun lowers and won't fire when you aim at the doors. Ghosts spawn outside and fly straight at you, and after hitting you they bounce back and attack again. **Normal Ghosts** (white) do damage. **Thief Ghosts** (purple) steal a random candy and flee, but shooting one inside the house gets the candy back. **Tank Ghosts** (big and dark) are slow, tough, and hit hard.

Health, strikes, and the clock
You have 100 health, which regenerates after a few seconds without being hit. A kid who runs out of time gives you a strike. The clock runs from 6 to 9 PM in four phases. Zero health or three strikes ends the night early.

5.3 Core Gameplay Loop



The whole game is this loop, with ghosts interrupting every step. It gets harder as orders grow, a second kid shows up, and the ghosts get stronger.

6. Game Modes

Twick o Tweat has the following game modes (according to Adams):

- **Primary gameplay mode, The Night (Serve & Defend):** first-person, dynamic camera, avatar-based interaction. All actions are available (move, look, shoot, interact). The UI is the full HUD: health, score, strikes, clock and phase, order cards, bag contents, power-up timers, and crosshair.
- **Tutorial mode:** same camera and controls, but only part of the gameplay is turned on. The clock is frozen at 6:00, and kids who time out don't give strikes. Step 1 has no timer and no ghosts. Step 2 has a timer and one 1-HP ghost.
- **Lights Out:** the rules are the same as the Night, but the presentation changes enough that it plays differently: you only see through a flashlight cone, and ghosts glow.

- **Shell menus (game paused):** the intro pop-up (instead of a title screen), information pop-ups at each new step or phase and the first color match, the pause menu (Resume, volume, Restart, Quit, high score), and the results / game over screen (score, high score, pumpkins if you won, Restart and Quit).
- **Competition mode:** single player against the game system, the only other competition is the saved high score.

Moving between modes: intro pop-up → tutorial step 1 (Start) → step 2 (one perfect delivery) → the Night (the tutorial kid is gone and the ghost is dead) → Lights Out (8:15 PM) → results (9:00 PM). Esc opens the pause menu from any gameplay mode. Zero health or a third strike jumps to Game Over, and Restart reloads a fresh night.

7. Rules

Following *Rules of Play*, the rules are split into **operational** (what the player is told), **constitutive** (the underlying math), and **implicit** (unwritten) rules. Together they should be enough to play the game without a computer.

7.1 Objective / Win Condition

Survive until 9:00 PM on the in-game clock. Surviving gives 1 pumpkin, a score of 5,000 or more gives 2 pumpkins, and 10,000 or more gives 3.

7.2 Lose / Setback Conditions

- **Health reaches 0** or **3 strikes** ends the night.
- **Setbacks:** wrong candy costs points, a timeout gives 0 points and a strike, thieves can take candy from a finished bag, and imperfect orders get no bonuses.

7.3 Operational Rules

- You hold one bag at a time, and grabbing a bag always gives you a new, empty one. Each E at a bowl adds one candy. The trash can empties the bag but you keep its color.
- Giving a kid the bag hands over everything in it, and you end up empty-handed. The next kid takes the empty spot right away.
- If a kid's timer runs out, they leave and you get a strike (not in the tutorial).
- Only perfect orders get bonuses. A perfect order in a matching color also gives that color's power-up.
- Normal and Tank Ghosts hurt you on contact, and Thief Ghosts steal one candy. Shooting a thief inside the house returns the candy.
- The gun can't fire at the front door, and health regenerates when you avoid hits.

7.4 Constitutive Rules

- **Orders:** the total candy count is random within the phase range, each candy is picked at random from the 5 types (repeats allowed), and the kid's color is random. Timer = 20 seconds + 6 seconds per candy.
- **Scoring per delivery:** +50 per correct candy, -25 per wrong or extra candy, 0 for missing candy. If the order is perfect, it also gets:
  - Speed bonus = 100 × (time left ÷ total time)

    - +50 perfect bonus
    - +100 if the bag color matches the kid
- A single delivery is never worth less than 0. Tutorial points count toward the final score.
- **Player:** 100 HP, and hits within 0.75 seconds of the last one are ignored. Regen is 4 HP per second after 3 seconds without being hit. Walk speed is 5 m/s. Each shot does 1 damage with a 0.15 second cooldown.
- **Power-ups:** last 5 seconds, each color has its own timer, and a repeat match restarts it. Light Blue sets speed to 8 m/s, Yellow sets damage to 2, Red heals to 100, and Green makes you invulnerable.
- **Ghosts:** spawn about 3 meters outside a random wall, only while fewer than the max are alive, and fly in a straight line at you through walls. Normal Ghosts deal 8 damage and get knocked back 3 meters. Thieves (30% of spawns from Phase 2) always have 2 HP and +1 m/s speed. Tanks (25% of spawns from Phase 3, one at a time) have 3× HP and 0.6× speed, and deal 15 damage. Every ghost is slower than the player.

Phase	Clock	Order size	Kids	Ghost types	Normal ghost HP / speed / spawn every / max
Tutorial	6:00 (frozen)	1 to 2	1	One 1-HP ghost (step 2)	n/a
1	6:00 to 6:45	2 to 3	1	Normal	2 / 2.0 m/s / 6 s / 3
2	6:45 to 7:30	2 to 4	2	+ Thief	3 / 2.2 m/s / 5.5 s / 4
3	7:30 to 8:15	3 to 5	2	+ Tank	3 / 2.4 m/s / 5 s / 4
4 (Lights Out)	8:15 to 9:00	4 to 6	2	All three	4 / 2.5 m/s / 4.5 s / 4

Each phase is 90 real seconds. Pausing and pop-ups stop the clock.

7.5 Implicit Rules

- **Don't shoot the kids:** enforced by the world, since the gun lowers at the door.
- **Stay in the house:** an invisible wall in the doorway blocks the player from leaving the map.
- **One night, one sitting:** there is no saving partway through. Closing the game changes nothing on your computer except the saved high score and volume.

8. Level Design & Progression

8.1 Level Overview

The game is one level in one room: the door and porch on two sides of the house, the candy table, bag table, and trash can. The tutorial teaches the order loop with no pressure, then adds the timer and shooting. After that, each phase adds one main new thing.

8.2 Using the Concepts in New Ways

- **Bag filling becomes a decision:** you learn it with 1 to 2 candies and any bag. Later, orders reach 4 to 6 candies, and the bag color decides which power-up you can earn.
- **One timer becomes two:** from Phase 2, you choose which kid to serve first and plan one trip that works for both.
- **Shooting changes purpose:** in the tutorial you shoot to protect your health. Thieves make you shoot to protect your **order**, the Tank makes you focus fire (or use Yellow), and Lights Out makes you find ghosts by their glow and sound before you can aim.
- **Health ties back into orders:** Red and Green bags turn order filling into healing and shielding.

8.3 Difficulty Curve & Playtime

Difficulty follows the clock, not your score: each phase raises the order size and the ghosts' health, speed, and spawn rate, and adds one new threat. The night is 6 minutes, and the whole game with the tutorial is about **7 to 8 minutes**.

9. Art, Animation & Audio

9.1 Visual Style

The room consists of basic shapes. The 3D models for kids, ghosts, candy bowls & bags, and the trash can were AI-generated. Ghost models were heavily inspired by Mario's Boo model, but I made sure it wasn't a blatant copy. A ghost's color tells you what it does, and a kid's color tells you which bag matches. The only hand-made art is the pixel-art sprites, which I drew myself.

9.2 Animations

There are no character animations. All motion is done in code: the gun tilts down at the door, ghosts bounce back with an ease-out after hitting you, trash can opening/closing, door opening, hits flash red, shots leave a tracer, and the lights flicker three times before the blackout.

9.3 Audio

Events	Source
Power-up, blaster shot, ghost hit	Sound effects I made in jsfxr
Background music, bag grab, trash, kid arrives, candy added, order success / fail, ghost enters, ghost death, thief steal, player hurt, tank growl, power outage, win, game over	Pixabay, under the Pixabay Content License
UI click	I recorded myself

No copyrighted material is used. Volume is controlled with the slider in the pause menu.

10. Iterations & Playtesting

10.1 Playtesting Process

For both playtesting rounds, I watched some people play and noted where they got lost, and I sent builds to others who sent feedback back. Each fix got its own "Playtest feedback:" commit.

10.2 Iteration Log

Commit	What changed	Why (playtest feedback / reasoning)
9f172d0	Phases shortened from 120 to 90 seconds	The night dragged, and with the tutorial it was close to 10 minutes
92af983	Thief Ghosts fixed at 2 HP	Late-night thieves were too tanky and always escaped
0a13038	Fixed tutorial timeouts giving strikes	Bug a tester ran into in the tutorial
097c965	Trash can moved between the tables, and added to the hint	Testers didn't notice it and handed over wrong bags
6924353	Door blocker and fall reset	A tester walked out the front door and off the map
065c9f9	HUD order timer is now a green-to-red bar	A number was hard to read at a glance
af49df4	Bag contents shown on the HUD	Testers lost track of what was in the bag
8f714a3	Color power-ups stack	A second match cancelled the first, which felt like a punishment
56274a8	Second kid moved to Phase 2	Phase 3 (two kids and the Tank at once) was a huge spike
afe260b	Ghost stats set per phase instead of by score, with slower spawns and less damage	Doing well made ghosts unpredictably stronger, and ghosts were too strong overall
a21f5c3	Hit cooldown and faster health regen	Ghost pile-ups drained health instantly
103e0bd	Audio added (v0.2)	Required, and testers got no warning when ghosts came from behind
30bfad1	Moved flashlight position	Gun model flickered white during lights out phase, so I re-positioned the flashlight to prevent it
b74b7c2	Updated score thresholds	Changed the thresholds for the 1-3 pumpkin score at the end of the game to better reflect the accomplishments

Commit	What changed	Why (playtest feedback / reasoning)
d1b92a7	Made ghosts spawn more often	With shorter phases and altered difficulty skew, the game became too easy, so I made the ghost spawn times a bit shorter to keep it hectic and engaging
78dca16	Moved the doors to opposite ends of the room and the tables to the middle	Playtester reported that placing the doors on opposite ends of the room and the tables in the middle would make the game more hectic but easier to keep track of the orders. I also added a door swing "animation" to door #2

10.3 Key Design Changes

Difficulty follows the clock instead of the score. Ghost stats used to go up every 500 points. That was a negative feedback loop that punished good players with rubber-banding they couldn't see. Now each phase has fixed stats, so the curve is the same for everyone and visible on the clock.

Power-ups stack and threats are spread out. Color matching is the game's positive feedback loop, so new matches shouldn't cancel old ones. Moving the second kid to Phase 2 means each phase adds one main new pattern, making sure that patterns come neither too fast nor too slow.

Updated pumpkin score thresholds. The score ranged drastically between players who got almost every order perfectly vs. those that made a lot of mistakes, but both got 3/3 pumpkins by the end. Updating the thresholds made good performance more rewarding.

Doors on opposite ends + tables in the middle. Having the tables next to the wall meant that ghosts that appear from the touching wall will catch the player off-guard, so moving them to the middle gave the player enough time to react to a ghost appearing. Moving the kids to opposite ends made it easier to tell apart which kid wants what, but at the same time made it more hectic since you have to move in more than one direction.

11. Credits & Assets

Design, programming, and sprites by me, Dani Zhumasheva. 3D Models for kids, ghosts, bowls, bags, and trash can were generated by Claude. Sound effects made in jsfxr, and music and other sounds from Pixabay (Pixabay Content License). Skybox image by Taplo, from OpenGameArt.org. Made in Unity 6.

12. Future Work

- **Endless mode** after beating the night, with no clock and difficulty that keeps climbing.
- **Real models and animations:** costumed kids who walk up and wave, and ghosts that look like ghosts.
- **Kid personalities and more candy:** impatient kids, picky kids, and more candy types.
- **Settings:** mouse sensitivity and separate music and sound effect volume.